

Reflection Paper: Data Analytics Internship at New Dia Cannabis Co.

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Company overview

New Dia Cannabis Co. operates in the heart of Boston's Fenway Park area and positions itself as the East Coast's First and Best cannabis mall. The 11,200-square-foot space blends retail, events, and experiential design to deliver an immersive shopping journey for a constantly changing audience. Foot traffic on Lansdowne Street shifts by the hour with Red Sox game days, concerts, and nightlife, which makes demand forecasting, merchandising, and staffing uniquely dynamic. The company's mission is to modernize cannabis retail through inclusive experiences, data-informed decisions, and high-velocity experimentation while navigating a heavily regulated environment. This combination of location, format, and regulatory complexity creates a living laboratory for customer analytics, pricing and promotions strategy, and operational excellence.

Role and activities performed

I served as an Data Analytics Intern and a strategic partner to New Dia's leadership team. My mandate was to architect an end-to-end growth intelligence stack that translated data into weekly operating targets, executive playbooks, and measurable commercial outcomes. The focus spanned pricing and promotions, retention, merchandising, game-day strategy, website optimization, and service design. I partnered cross-functionally with marketing, inventory, store operations, vendors, and investors, and translated analytical outputs into floor-ready actions.

Weekly strategy and decision support

I maintained a weekly insights cadence to support upper management with variance bridges, forecasts, and tactical adjustments. The work covered traffic and ID scans, conversion, units per ticket, average ticket value, gross margin dollars, new-versus-returning mix, and program-level attribution. I reconciled discrepancies across dashboards and sources so the same metric meant the same thing everywhere, which raised decision velocity and trust.

Campaigns, discounts, and experimentation

I helped plan discounts and activation calendars, and I designed multi-armed bandit tests to allocate promo exposure toward higher-ROI variants. I conceptualized baseball-themed offers aligned to the MLB schedule, created day and hour leaderboards by category, and implemented

profitability guardrails. During a Blue Jays week initiative, I isolated likely-fan demand using time-window filters and source logic, narrowing 49,432 rows to a 27,639-ticket, 51,521-item fan sample for lift comparisons versus baseline. I quantified attach rates, cross-sell opportunities, and margin impacts to steer the final menu and operational playbook.

I supported Pledge Dia, the flagship student-focused acquisition and loyalty program that generated more than 5,000 new customers in a single month in 2024, and scaled bigger in 2025. I contributed segmentation, channel mix strategy, CAC-to-payback CLV modeling, redemption propensity reads, and a measurement plan that tied creative and offer depth to actual net sales and repeat behavior.

Retention, loyalty, and service design

I mapped loyalty touch points across the journey and analyzed first-to-repeat dynamics over 180 days using survival-style reads. I proposed a POS-based NPS capture to establish a continuous CX signal and linked staff-assist behaviors to conversion. I designed and ran an exit-survey experiment with clean controls, t-tests, effect sizes, and confidence intervals, which revealed large conversion deltas for assisted customers. These findings were operationalized through staff training, signage, and incentive design.

Merchandising, elasticity, and portfolio strategy

I integrated POS, line-item, discount, inventory, and customer data to build log-log OLS price elasticity reads at SKU and category levels. I conducted association-rule mining to identify reliable cross-sell ladders, with special focus on bulk flower and accessories. I delivered an accessory category audit with YTD KPIs, attach opportunities, and forecast scenarios. I executed portfolio clustering for 125 SKUs using k-means on eight standardized attributes, achieving a silhouette near 0.61, and codified role-based cluster naming and attach playbooks.

Forecasting and scenario planning

I developed six- and nine-week forecast scenarios for apparel and built an attendance-to-revenue model to support a Fenway blueprint targeting seven-figure seasonal throughput. I layered capacity constraints, staffing requirements, and gross margin implications to make the scenarios execution-ready. For playoffs, I modeled traffic and product mix shifts and built action triggers that linked live KPIs to inventory and staffing decision trees.

Analytics engineering and internal software

To remove manual friction, I developed an automated attendance tracker using AI, that parses scheduled shifts and time-stamped punches, classifies tardies and early departures, and computes attendance by hours plus average break times. This tool automated days of manual manager work each month. I also built SQL-powered dashboards on casa quicksight powered by AWS, and helped standardize a canonical KPI layer across CASA by QuickSight, Alpine IQ,

GA4, Google Ads, Dutchie, Happy Buyers, and Surfside. I resolved cross-dashboard inconsistencies, added discount name and group fields to transaction tables, and ensured that totals and derived KPIs reconciled across filters and time horizons.

Channel, web, and field enablement

I partnered with CannaPlanners and Serpcom to audit website flows and paid channel funnel metrics. I flagged product-to-checkout drop-offs, rationalized high-intent keyword clusters, and surfaced content gaps for blog and SEO. I designed weekly and monthly goal frameworks for sales reps and management, ran weekly briefs, delivered monthly executive presentations, and provided procurement signals for aging inventory. I also introduced non-buyer data capture to measure consideration without purchase, which improved read-through on conversion rate and offer effectiveness.

Tool and platform stack

I worked hands-on with CASA by QuickSight for BI, Alpine IQ for segmentation and loyalty data, GA4 and Google Ads for web and performance analytics, CannaPlanners for site monitoring and content, Dutchie for POS and menu, Happy Buyers for inventory aging, Surfside for media and audiences, SparkPlug for incentives, Serpcom for search and social ad analytics, Google big query to run queries on cloud, python via google colab, tableau to display reports, spss for statistical modelling, cursor AI for automating my ad hocs and emails via multi agentic framework and combined ID scans with CCTV logs to understand hotspots and conversion rates.

Skills and experiences acquired, and how they enhanced MSBA strengths

Analytical depth in causal and predictive methods

I operationalized survival analysis for first-to-repeat timing, which sharpened my cohort thinking from the Customer Analytics coursework. Survival curves, hazard-like reads, and graduation ladders informed offer depth for new versus returning customers. Logistic propensity-to-redeem models improved promo targeting and reduced wasted discounting by moving budget toward higher-lift microsegments. Log-log OLS elasticity estimates supported pricing decisions that balanced gross margin with demand, converting theoretical lectures on elasticity into practical, SKU-level guidance.

Segmentation, mining, and portfolio strategy

MSBA modules on clustering and dimensionality reduction translated directly into the k-means portfolio project. Standardizing eight attributes and interpreting a silhouette near 0.61 taught me to balance statistical fit with business interpretability. I learned to translate clusters into roles like

entry, core, premium, and halo, then to write attach playbooks that guided team on how to graduate customers across tiers. Association-rule mining reinforced the need to test attach logic against support, confidence, and lift, and to select rules that scale operationally with signage and training.

Promo and price architecture

I strengthened discipline in the economics of promotion. I instituted non-stack guardrails, GM per discount dollar reporting, and Net Sales per Promo Dollar as north-star metrics. I measured promo penetration, leakage, and stacking, and constructed comparisons that held product mix and timing constant. The multi-armed bandit framework connected my MSBA understanding of exploration-exploitation tradeoffs with real campaign decisions, letting us learn faster while capping downside risk. CAC-to-payback CLV modeling aligned creative and offer depth to acquisition quality, not just top-line lift.

Experiment design and measurement

Working on the staff-assist study allowed me to run a true field experiment. I operationalized clean control construction, ran t-tests with effect sizes and confidence intervals, and reported intent-to-treat style reads to avoid bias from incomplete compliance. This cemented my MSBA foundations in causal inference and reinforced the importance of instrumenting for operationalization, not only statistical significance. I also learned to pre-plan power where feasible, especially on shorter, high-variance tests tied to event days.

Forecasting and operations integration

Scenario planning became a core competency. I built baselines, layered lever-uplift assumptions, stress-tested against temporal volatility, and turned the outputs into staffing and inventory guidance. I learned to translate attendance math into revenue and margin forecasts while honoring capacity constraints. This directly linked my MSBA forecasting coursework to labor planning, assortment prioritization, and procurement.

Commerce analytics engineering and data governance

Cleaning and reconciling cross-dashboard discrepancies taught me to build from a canonical KPI layer. I aligned KPI definitions across CASA by QuickSight, Alpine IQ, Dutchie, and ad platforms so leadership could trust every dashboard and quickly resolve variance. I added missing fields like discount names and groups so transactional tables I built that could support more nuanced analysis of promo performance and margin.

CX instrumentation and website analytics

The web funnel audit drew directly on Digital Marketing Analytics training. GA4 exploration surfaced product-to-checkout drop-offs and session quality patterns. I partnered with the web

team on content, internal linking, and high-intent keywords, and I proposed POS-based NPS to create a continuous CX telemetry loop that bridges in-store and online experience. This infused MSBA measurement frameworks into an omnichannel retail context.

Executive enablement and cross-functional leadership

I translated outputs into decision accelerators. Each weekly brief included variance bridges, root-cause reads, and recommended actions. Executive playbooks codified promo guardrails, attach ladders, and survival-based offer depth for specific cohorts. I worked with SparkPlug to align incentives to behavioral targets such as loyalty sign-ups, attach of under-penetrated categories, and graduation from entry to core SKUs. I met with vendors to shape co-op offers that met our guardrails. I learned to communicate in a way that balanced statistical rigor with operational simplicity.

Ethical, regulatory, and practical judgment

Operating in a regulated category honed my respect for compliance, privacy, and responsible advertising. I designed measurement that avoids personally identifiable misuse and supported professional standards in reporting and data storage. Practical judgment mattered as much as modeling accuracy. On game days, a clean rule of thumb that staff can apply beats a perfect model that no one can run at 6 p.m.

Concrete outcomes and personal growth

Key outcomes included scaling a flagship student program with a strong measurement backbone, upgrading the fidelity and consistency of KPIs across systems, delivering a KPI goal for all employees to work together and trained them to get them familiar on how much can we sell, improving attach strategies with portfolio clustering, and proving service impact through the staff-assist experiment. I learned to be comfortable in ambiguity, to start with the clearest useful measure, and to layer sophistication only when it yields incremental value. I also learned that great analytics are social. Sitting with budtenders, listening to the inventory team's constraints, and aligning with marketing and web teams made the work stick.

How the internship enhanced my MSBA strengths

1. Customer Analytics
Cohorts and survival-style reads turned classroom concepts into weekly practice. First-to-repeat timing, graduation ladders, and loyalty touch points were embedded into the way we sized promo impact and forecasted repeat revenue. This sharpened my CLV mindset and grounded acquisition planning in lifecycle economics.
2. Causal and Statistical Methods
I applied t-tests, effect sizes, and confidence intervals in a live retail environment and learned to design intent-to-treat reads when compliance is imperfect. I used logistic

propensity modeling for offer targeting and survival-like analyses for time-to-repeat. The experience improved my ability to separate signal from noise in noisy, fast-moving data.

3. Pricing and Revenue Optimization

Elasticity estimates, margin guardrails, and promo architecture aligned directly with coursework in pricing. I learned to select price changes where elasticity supported the margin objective and to use offer depth as a controlled lever with clear profit caps.

4. Predictive Modeling and Segmentation

K-means clustering for role-based SKU strategy, redemption propensity scores, and association rules translated MSBA predictive techniques into choices on assortment, signage, and staff prompts. I focused on models that staff could act on in real time.

5. Forecasting and Operations

I integrated forecasts with staffing, procurement, and capacity constraints. I learned to design action triggers and to communicate uncertainty honestly, which built trust and improved outcomes when actuals diverged from plan.

6. Measurement and KPI Design

Building a canonical KPI layer and reconciling dashboards strengthened my measurement systems thinking. Knowing exactly how each metric is defined and ensuring that totals and derived KPIs reconcile across tools is the backbone of executive trust.

7. Data Engineering and Automation

The attendance generator and SQL-powered dashboards grew my fluency with messy data, schema drift, and resilient transformation logic. I now approach analytics as a product that must be reliable, maintainable, and extensible.

8. Communication and Storytelling

Weekly briefs and monthly executive presentations honed my ability to frame problems, quantify drivers, and recommend actions in clear language. I learned to keep charts simple, call out what changed and why, and show the operational next step.

Conclusion

This internship was a full immersion in analytics-driven retail. I integrated advanced modeling with live operations, and I learned to convert statistical outputs into frontline decisions. I built systems that scale, and I validated strategies through experiments and scenario planning. The experience deepened every core MSBA competency, from customer analytics to hype marketing and causal inference to pricing, forecasting, and executive communication. Most importantly, it reinforced that analytics only creates value when it is trusted, operational, and in rhythm with the people who use it. By the end of the summer, I had moved from reporting what

happened to engineering to how we decide, and I leave with a playbook and a mindset that I can apply to any data-rich, fast-moving business.

THANK YOU